

Problem 1 (§10, #2). Prove that for every positive integer n ,

$$\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}.$$

Problem 2 (§10, #4). Prove that for every positive integer n ,

$$\sum_{k=1}^n k(k+1) = \frac{n(n+1)(n+2)}{3}.$$

Problem 3 (§10, #8). Prove that for every positive integer n ,

$$\sum_{k=1}^n \frac{k}{(k+1)!} = 1 - \frac{1}{(n+1)!}.$$

Problem 4 (§10, #12). Prove that $9 \mid (4^{3n} + 8)$ for every integer $n \geq 0$.

Problem 5 (§10, #16). Prove that for every positive integer n , $2^n + 1 \leq 3^n$.

Problem 6 (§10, #18). Suppose $\{A_k\}_{k \in \mathbb{Z}}$ is a family of sets in some universal set U , and $n \geq 2$. Prove that

$$\overline{\bigcup_{k=1}^n A_k} = \bigcap_{k=1}^n \overline{A_k}.$$

Problem 7 (§10, #26). Let F_n be the Fibonacci sequence. Prove that

$$\sum_{k=1}^n F_k^2 = F_n F_{n+1}.$$

Problem 8 (§10, #42). Prove that the n th Fibonacci number F_n is even if and only if $3 \mid n$.